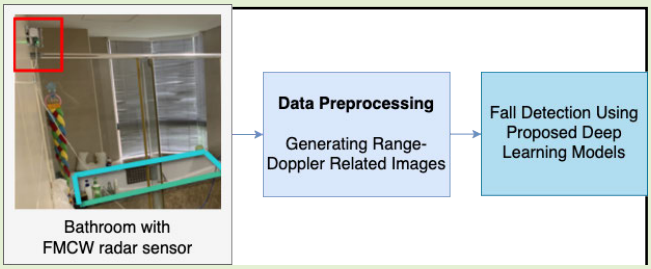
A Fall Detection System Based on FMCW Radar Range-Doppler Image and Bi-LSTM Deep Learning

Wei-Lun Hsu, Jin-Xian Liu, Chia-Cheng Yang, and Jenq-Shiou Leu, *Senior Member, IEEE*

***Abstract*—Fall detection has become a popular research topic in recent years due to the fact that falls can cause serious injuries. Most existing fall detection systems rely on wearable devices or image recognition technology. However, these approaches possess limitations. For instance, most people take off wearable devices during showers, while cameras with image recognition raise privacy concerns. To overcome these challenges, this study proposes a novel approach using 60-GHz radar to develop a fall detection system. We collected raw data from the frequency-modulated continuous wave (FMCW) radar in three bathrooms and two other indoor areas. Then, we derived features by generating range-Doppler images and proposed a novel preprocessing method, which can generate a range-Doppler histogram with a longer time interval. In addition, we proposed two deep learning models, a trigger model and a fall detection model, for the fall detection system. The trigger model employs convolutional neural networks (CNNs) to identify range-Doppler images with significant speed and distance variations and is optimized to run on devices with limited central processing unit (CPU) resources. The fall detection model, utilizing a bidirectional long short-term memory-based deep learning model, aims to reduce the possibility of false alarms and make the final decision. Over 700 records were collected during the experiment, and the fall detection system achieved an accuracy rate of nearly 96%, with recall and precision rates exceeding 90%.**



***Index Terms*—Deep learning, fall detection, frequency-modulated continuous wave (FMCW) radar, long short-term memory (LSTM), range-Doppler.**

I. Introduction

FALLS may cause significant harm. Recently, fall detection has become a popular research topic due to the prevalence of slips, falls, and fainting in places, such as bathrooms, homes, outdoors, and hospital wards. Irreparable injuries may occur if the victim is not discovered in time and rescue efforts are delayed.

To detect falls promptly, numerous pieces of equipment and technologies are widely used in fall detection nowadays. The most popular ones are wearable devices, such as the Apple Watch [1], and fall detection systems that utilize cameras for image recognition [2]. However, our extensive study revealed various issues. First, smart wristbands may be equipped with sensors, such as accelerometers and gyroscopes, to detect falls. The challenge is that they need to be worn and have water-resistant technology; however, most people tend to take them off while showering. Second, although fall detection based on camera image recognition has high accuracy, the issue of privacy rights is a concern for many people.

The sensors that can overcome these two issues are millimeter-wave [3] and infrared (IR) sensors. Radar waves and microwave sensors operate based on the Doppler effect [4] of electromagnetic waves. Radar and microwave sensors are highly sensitive, penetrating, and diffuse, making them effective in avoiding obstacles and suitable for open and complex environments [5]. Infrared sensing is the use of infrared rays emitted by the human body or heat sources to detect moving objects [6]. However, in complex environments, such as bathrooms, there may be temperature variations and interferences that could affect the sensitivity of infrared sensing and make it unsuitable for detecting falls.

Jimi et al. [3] proposed a bathroom monitoring system using a 79-GHz millimeter-wave ultrawideband sensor with a

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fast chirp modulation (FCM) scheme and a hidden Markov model (HMM). However, the system was only tested in a bathroom, lacking adaptability to different environments. Therefore, we present a novel fall detection system that is adaptable to various environments and highly accurate. The experiments in this study not only included testing in three different bathroom environments but also in an office and a conference room to demonstrate the system's versatility.

Fall detection systems based on radar face the challenge of reducing false alarm rates [7]. In this study, we propose a novel range-Doppler image preprocessing method and develop two deep learning models for a two-stage detection process. Our approach not only addresses the issue of computational resource requirements but also effectively lowers the false alarm rate. By implementing these advancements, we aim to improve the accuracy and reliability of radar-based fall detection systems, leading to increased safety and security for individuals in various environments.

Based on the above, the main contributions of this article are as follows.

1) We developed a fall detection system that operates without field constraints while achieving a high degree of accuracy within a specific range.

2) We proposed a novel detection mechanism for two radar-based fall detection models that utilize deep learning technology, resulting in a reduction in computational resources.

3) We proposed a novel range-Doppler image preprocessing method, which increases accuracy and reduces possibility of false alarms in the fall detection system.

The remainder of this article is organized as follows: Section II provides a brief background and the summary of recent studies on fall detection. Section III gives a brief overview of the fall detection system architecture, the novel radar data preprocessing methods, and the proposed deep learning models. Section IV evaluates the performance of the proposed fall detection system and presents the experimental results. Finally, we conclude this article in Section V.

II. Related Work

In recent years, advancements in technology have transformed fall detection into a solvable problem, thereby drawing considerable attention from researchers. Lu and Chu [8] introduced the image-based fall detection system (IFADS). However, this system is inapplicable in bathroom settings due to potential privacy issues associated with the use of image-based detection. Numerous studies have delved into the use of wearable devices for fall detection [9]. However, these devices pose certain challenges when employed in bathroom environments, as they necessitate the use of waterproof technology and may encounter resistance from users unwilling to wear such devices in this setting.

In an attempt to address these challenges, numerous studies have utilized radar technology for fall detection. Liu et al. [10] developed a Doppler radar-based fall detection system that represents the Doppler signatures of various human activities using mel frequency cepstral coefficients (MFCC). They employed support vector machines (SVMs) and k-nearest neighbors (kNNs) to identify falls. Nevertheless, in recent years, beyond traditional classification algorithms, such as SVM and kNN, a significant number of studies have harnessed the potential of deep learning techniques for fall detection. Convolutional neural networks (CNNs) and long short-term memory (LSTM) models are the examples of these techniques and have been applied in several fall detection studies [11].

Wang et al. [12] introduced a fall detection method using millimeter-wave radar, employing a line kernel CNN (LKCNN). Despite incorporating deep learning techniques and testing their approach in complex environments, the performance of their system in bathroom settings remains unexplored. On the other hand, Yang et al. [13] proposed a radar-based fall detection system leveraging a CNN model for fall recognition. Their study demonstrated that simultaneous usage of two radars could enhance the system's accuracy. However, they did not provide additional metrics to reflect the system's rate of false alarms.

In radar-based fall detection research, both CNN [14] and LSTM [15] models have been extensively utilized. CNN, renowned for its proficiency in feature extraction [16], has proven to be highly efficient in extracting features from range-Doppler image. Given that a fall represents a continuous action, an LSTM model, skilled at learning long-term dependencies in data sequences, can be particularly effective in identifying falls [17]. While most fall detection studies rely on the judgment of a single model, our research proposes a novel detection mechanism, dual-model approach. The first model, utilizing CNN, is designed to conserve computational resources by isolating only those range-Doppler images that exhibit significant changes in speed and distance. The second model employs LSTM to consider continuous action and utilizing our proposed novel preprocessing histogram to mitigate the occurrence of false alarms.

Drawing from the strengths and weaknesses identified in the reviewed literature, this study proposes a fall detection system that navigates space limitations. By implementing the proposed preprocessing methods, models, and a novel fall detection workflow, we can significantly optimize computational resources and reduce the probability of false alarms, thereby resulting in a more reliable fall detection system.

III. Methodology

The flowchart of the fall detection system is shown in Fig. 1. In this section, initially, we elaborate on the methods employed for data preparation and preprocessing, which involve the range-Doppler image and long-time interval range-Doppler histogram. Following this, we delineate the process and methodologies adopted for data collection and labeling. Lastly, we introduce the architectures of the two proposed deep learning models.

A. Generating Range-Doppler Image

Given the irregularity of falling movements, a fixed and consistent action cannot be reliably used as a recognition feature, as is often the case in gesture recognition. The feature of directionality is also not applicable under such circumstances.

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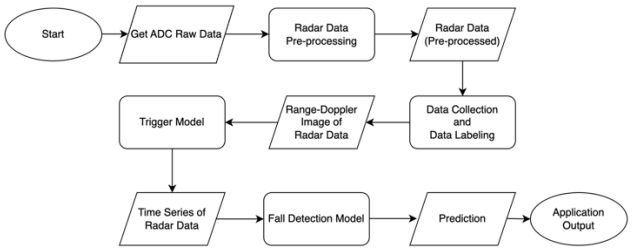

Fig. 1. Flowchart of the fall detection system.

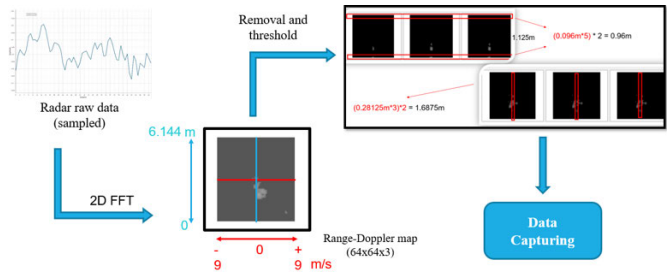

Fig. 2. Preprocessing flowchart for the range-Doppler image.

Consequently, we propose a feature that integrates changes in distance and speed for recognition purposes.

Upon receiving a single frequency modulated continuous wave (FMCW) radar signal [18], often referred to as a chirp or a fast time signal, the signal is sampled and subjected to analysis. The fast Fourier transform (FFT) is then applied to convert the signal from time-domain representation to frequency domain. By analyzing the beat frequency of each chirp, it is possible to calculate the relative distance.

For range resolution to be dependent on frequency bandwidth, the formula is as follows:

$$d_{\text{res}} = \frac{C}{2B}. \tag{1}$$

In terms of max range, the formula is as follows:

$$d_{\max} = \frac{f_s * C}{2S} = \frac{C * N_s}{2B} \tag{2}$$

where C represents the speed of light, B represents the bandwidth, f_s is the sampling frequency, S represents the slope, and after multiple transformations, N_s represents the number of samples, and T_c represents the time of the triangular wave.

A series of transmit (TX) signals, typically observed over a period of time rather than as a single chirp, are generated at equal intervals. These signals are known as frames or long-time signals. Assuming that there are n TX signals at equal intervals, each corresponding intermediate frequency (IF) signal will have a peak at the same position post Fourier transform (FT). The respective discrete signals consist of two phases that rotate at frequencies ω_1 and ω_2. As a result, this discrete signal will exhibit two peaks following the FFT, corresponding to frequencies ω_1 and ω_2. The frequency ω can then be utilized to ascertain the respective speeds using the formula $v = (\lambda\omega/4\pi T_c)$.

In terms of velocity resolution, the formula is as follows:

$$V_{\text{res}} = \frac{\lambda}{2T_f}. \tag{3}$$

Max velocity can be calculated using the following formula:

$$V_{\max} = \frac{\lambda}{4T_c} \tag{4}$$

where λ represents the wavelength of the radar signal, T_f represents the frame time, and T_c represents the time of the triangular wave.

Fig. 2 illustrates the use of a radar with one transmitter and three receivers, resulting in the display of three signal images from the received data. The image on the left side depicts the raw data, while the right side presents the transformed range-Doppler images, each frame of which represents a time duration of 33 ms. The x-axis of these images represents speed, and the y-axis represents distance (range).

B. Method for Range-Doppler Image Cutting

Based on the parameters we set for the radar, the distance of the range-Doppler varies between 0 and 6.144 m, and the speed variation can reach up to 9 m/s. The radar is positioned at a height of approximately 2 m, and it is placed at an angle in one of the four corners of the field.

We have noticed that the distances shown on the range-Doppler image are not clearly discernible. For example, as shown in Fig. 3, at sections marked "1" and "2" in red, if a person is located too close to the radar, their presence may not be detected unless their height is very close to the radar. Furthermore, some of the readings may be attributed to noise reflected by objects in the corner. The changes in velocity observed at the section marked "3" in Fig. 3 are a natural result of a person's movement. To reduce image complexity and facilitate the identification of a person's falling speed and effective range, we have to perform partial cuts on the range-Doppler image.

In Fig. 3, the size of each image is 64 × 64 pixels, with the x-axis representing velocity and the y-axis representing range. Given that the radar detection range is set at 6.144 m, we removed the sections marked "1" and "2" in Fig. 3, which together measure 0.96 m (0.096 m × 5 rows × 2), to reduce image complexity. Additionally, we removed the section marked "3" in Fig. 3, which spans six columns. Each column represents a velocity change of ±0.28125 m/s, resulting in a removed range with a velocity change of 1.6875 m/s (0.28125 m/s × 3 columns × 2 directions). The image size after processing is 54 × 58 pixels.

C. Long-Time Interval Range-Doppler Histogram

The use of range-Doppler images for discerning between individuals in the field can be challenging. To address this limitation, a novel approach is proposed using a range-Doppler histogram with a longer time interval, as shown in Fig. 4. The histogram extends over a time span of 32 frames, with each frame representing 33 ms. This longer time interval provides more comprehensive information.

In the process of generating a histogram, only the first chirp from each frame is considered. After collecting the first chirps from 32 consecutive frames, an FFT is performed to generate a range-Doppler image. This image allows for the extraction of

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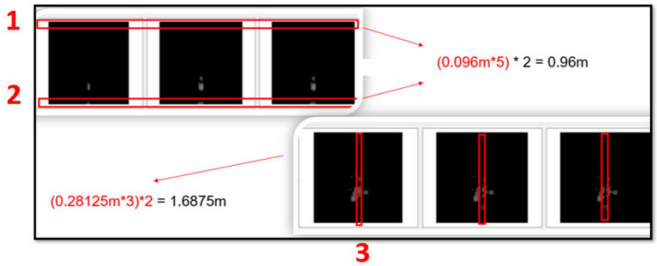

Fig. 3. Diagram of range-Doppler image cutting.

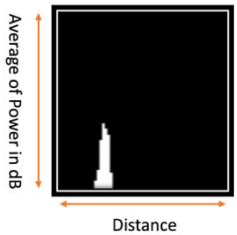

Fig. 4. Proposed long-time interval range-Doppler histogram.

information on range, velocity, and power (in dB). The power values (in dB) within each range bin are then summed and averaged. Subsequently, these averaged values for each range bin are normalized between 0 and 1. This results in what we refer to as the "average of normalized power (dB)" for each range bin in the range-Doppler image. The resulting histogram represents the distance (on the x-axis) against the average of normalized power (dB) (on the y-axis).

The input to the model is a 1-D array, where each element represents the "average of normalized power (dB)" for a specific range bin. However, the histogram shown in Fig. 4 is designed for data labeling and visualization purposes. To achieve this, the "average of power in dB" values within this histogram is scaled and normalized within the range of 0–254.

By lengthening the image interval and focusing on the first chirps of each frame, the proposed histogram enhances the ability to differentiate between individuals. It can reduce the possibility of false alarms caused by significant changes in range-Doppler image, especially in scenarios with multiple moving individuals. The complexity of the range-Doppler image in such cases can affect the performance of the fall detection system. However, by considering continuous frames in the proposed histogram, the movement patterns of multiple people can be distinguished from a fall, thereby reducing the occurrence of false alarms.

Another advantage of the proposed histogram is its ability to filter out data that are too far away for effective radar detection. As the radar is installed at a high place, the detected power decreases during a fall due to increased distance. The histogram can be used to exclude data that are beyond the radar's effective detection range during the data collection and labeling process.

D. Data Collection and Labeling

Data collection was conducted in three bathrooms of varying sizes, along with one office and one conference room. The radar was placed at a height of approximately 2 meters, and tilted at an angle of about 15 degrees. The process flowchart is depicted in Fig. 5. At this stage, our primary objective is to gather, analyze, and label the available data features for model training.

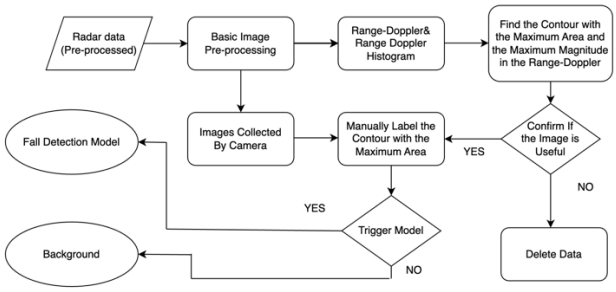

Fig. 5. Flowchart of data collection.

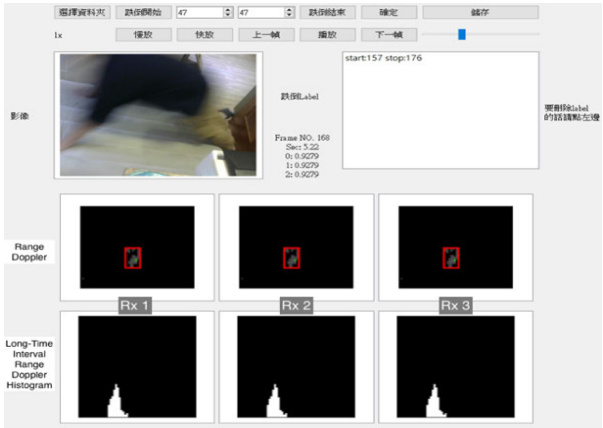

Fig. 6. Tool for data labeling.

Our data collection primarily focuses on acquiring raw radar data and camera images. In order to monitor changes in the radar data, we transform it into Range-Doppler images, which are then used as the basis for subsequent labeling. As the frame rates of the radar and camera differ, they cannot be synchronized. To compensate for this, we assign timestamps to both the radar and camera images.

We developed a tool using Python-pyqt5 for the purpose of data labeling. This tool allows us to verify the quality of both camera and range-Doppler images, as shown in Fig. 6. We use this tool to label falls based on both types of images. Typically, if a fall occurs, significant changes in speed will be evident in the range-Doppler image. Each frame of the image is then labeled with either a 0 or 1, indicating the absence or presence of a fall, respectively.

E. Deep Learning Model for the Fall Detection System

This study proposes two deep learning models for a fall detection system: the trigger model and the fall detection model. The trigger model is designed to recognize range-Doppler images with significant speed and distance variations and is optimized to reduce its model size, enabling deployment on devices with limited computational resources. The fall detection model is designed to reduce the probability of false alarms and is responsible for making the final decision on whether a fall has occurred.

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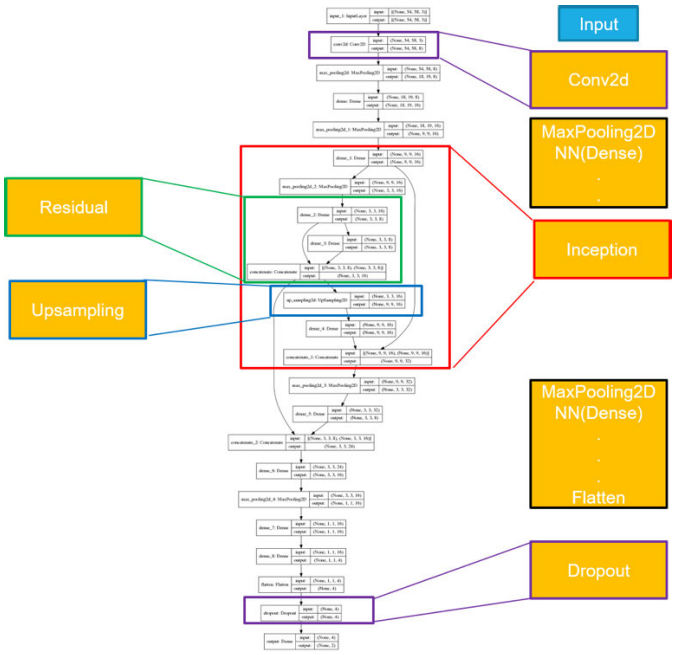

Fig. 7. Architecture of the proposed trigger model.

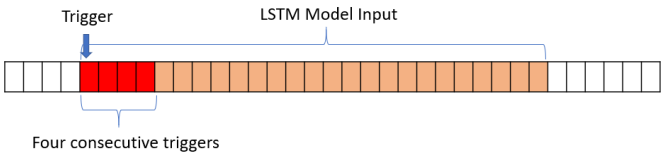

Fig. 8. Schematic of the trigger algorithm.

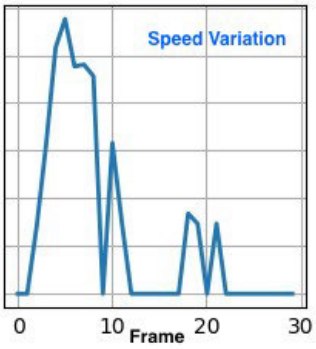

Fig. 9. Illustration of the velocity change during a fall.

F. Architecture of the Proposed

The trigger model is employed to filter range-Doppler images, thereby eliminating behaviors that do not exhibit significant speed changes. The fall detection model is then activated only when the range-Doppler image exhibits clear speed changes, upon which it performs more advanced predictions. This mechanism optimizes computational resources and enhances the accuracy of the fall detection system.

Fig. 7 shows the architecture of the trigger model. The input of the model consists of frames from each of the three receivers, which are range-Doppler images after being cut.

To address the noise in radar data, the proposed trigger model leverages a CNN [19] as a filter to enhance the image clarity. The next stage uses max-pooling [19] to retain significant features while reducing the image size for faster computation. However, as the image features get compressed, the emphasis on certain features becomes less noticeable. To solve this problem, the model adopts upsampling [19] technology and a modified version of the inception [20] architecture to increase the feature width, capture diverse features, and avoid feature bottlenecks. In order to enhance feature extraction, the model implements a residual architecture [20], [21] that allows lower level features to merge with higher level features. The leaky rectified linear unit (ReLU) [22] is primarily used as the activation function. The model also uses flatten to transform dimensions and employs dropout [23] to prevent overfitting [24]. The loss function used is binary cross-entropy.

G. Proposed Fall Detection Model

Based on the prediction of the trigger model, the data input to the fall detection model mainly consists of information with large speed variations in the range-Doppler image. In this model, it is not only necessary to improve the accuracy of the fall detection system but also to minimize the probability of false alarms.

The input data for the trigger model are single-frame range-Doppler image. However, the fall detection model is mainly used to identify the continuous action of falling, which requires considering multiple frames. Therefore, the proposed fall detection model is designed based on bidirectional LSTM (Bi-LSTM). In addition, we must select an appropriate continuous data length for the model.

To enhance the accuracy of the model, we have designed a more precise trigger mechanism. Since falling is a continuous action, the trigger mechanism functions by detecting four consecutive frames as falling using the trigger model, as shown in Fig. 8. These n consecutive frames are then utilized as features for the fall detection model, with the value of n determined through several experiments. Based on Fig. 9, a majority of frames during the falling period can be classified as falling; thus, the optimal number of frames for the model is 25.

The proposed fall detection model comprises three components, including a fusion module that integrates the long-time interval range-Doppler histogram with the range-Doppler image. The remaining two components separately extract high-level features from the long-time interval range-Doppler histogram and range-Doppler image, which are then combined for the final output. The architecture of the model is depicted in Fig. 10. In order to achieve the linear interpolation effect, a CNN model is utilized to downscale the range-Doppler image. To further enhance the model, we employ the inception architecture [20] to divide it into smaller models that can produce their own features and expand its width. To incorporate long-time interval range-Doppler histogram features, a residual architecture [21] is used to connect lower level features with higher level features, which can enhance the features. In addition, the accuracy of the model is increased by utilizing the Bi-LSTM [25].

To effectively capture features from both long-time interval range-Doppler histogram and range-Doppler image, we first

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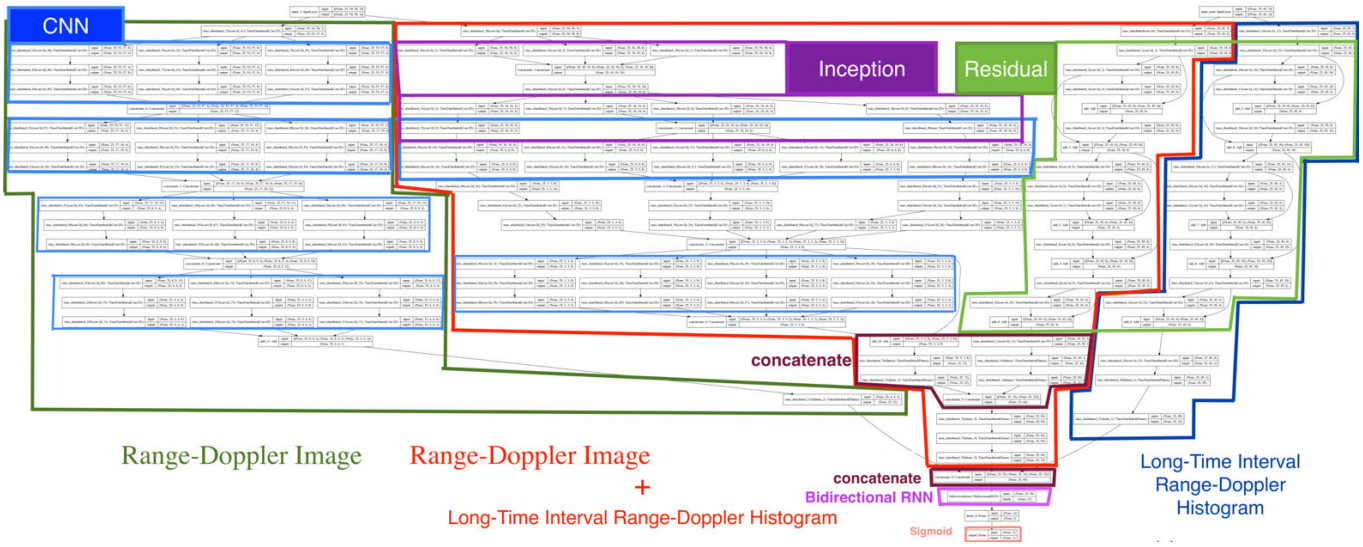

Fig. 10. Architecture of the proposed fall detection model.

TABLE I
INFORMATION FUNDAMENTAL TO THE RADAR SENSOR

Device Name	Infineon XENSIV DEMO 60GHz TR13CAIP
Tx/Rx numbers	Tx:1/Rx:3
Max Range	15m
Carrier	60GHz
Signal	FMCW
Bandwidth	5GHz
Range Resolution	down to ~3cm

concatenate them to produce a feature set. To ensure the distinctiveness of the features, each feature set is subjected to individual feature extraction until they are merged using the concatenate function. Finally, the output is assessed for similarity using the sigmoid function.

IV. EXPERIMENT

A. Experimental Device

The demo board used in this research is the TR13CAIP, and the radar is BGT60TR13C [26], which is equipped with one transmitting antenna and three receiving antennas, ranging in frequency from 58 to 63 GHz, with a bandwidth of 5 GHz and a central microcontrol unit, as shown in Table I.

B. Experimental Field

In order to allow the fall detection system to be deployed in a variety of environments, we have chosen five experimental fields, including an office, a meeting room, and three bathrooms of different sizes. One of the bathrooms is shown in Fig. 11. In the experiment, we set up the radar at a height of about 2 m and placed it at an inclination of about 15°. The horizontal angle is based on the experimental observation and relevant literature, which show that the radar emits from −60° to +60°, which is relatively far from the received energy and is more obvious. Hence, the radar set up in one of the four corners of the field can observe the range of clear energy most effectively.

Fig. 11. Image of one of the experimental fields.

C. Introduction to the Dataset

The study participants consisted of 11 male and 3 female individuals of Asian descent. The age range was between 22 and 25 years old, with male heights ranging from 166.5 to 183 cm and an average height of approximately 171 cm. Male weights ranged from 57 to 88 kg with an average weight of 68.9 kg. The average height and weight for females were 160 cm and 46.6 kg, respectively. There are numerous types of falls, as referenced in [27] and [28]. However, in the experiments conducted in this study, we mainly focused on simulating instantaneous falls because this type of fall is typically more dangerous.

The proposed fall detection system comprises two deep learning models: the trigger model and the fall detection model. Therefore, data analysis plays a crucial role. For comparative purposes, we divided all the collected data into two datasets. There is minimal variance between the two datasets, with the majority of the differences arising from the number of nonfalls, as the data on falls remain consistent.

1) Dataset A: Dataset A is the complete dataset of falls, including data on a variety of nonfalls, common behavior in the bathroom, and situations that easily lead to false alarm, as shown in Table II.

2) Dataset B: The dataset has been improved to resemble Dataset A by removing nonfall data that could potentially lead

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TABLE II
OVERVIEW OF DATASET A

Fall Data		**Non-Fall Data**	
Field	Number	Action	Number
Office	346	Squat down, Sit down, Stand up	2
Meeting Room	66	Walk	48
Bathroom 1	43	Wave hand	30
Bathroom 2	40	Brush teeth	15
Bathroom 3	40	Take a shower	19
Others	26	Hold the shower	17
		Walk through the door	57
		Others	10
Total	561	Total	198
Label Frame Size: 17~24 Frames (FPS: 30)			
Number of Total Records: 759			

TABLE III
OVERVIEW OF DATASET B

Fall Data		**Non-Fall Data**	
Field	Number	Action	Number
Office	346	Squat down, Sit down, Stand up	2
Meeting Room	66	Brush teeth	15
Bathroom 1	43	Walk through the door	12
Bathroom 2	40	Others	3
Bathroom 3	40		
Others	26		
Total	561	Total	32
Label Frame Size: 17~24 Frames (FPS: 30)			
Number of Total Records: 593			

to false alarms. The examples of such data, including lotus head flushing, showers, and other potential sources of false alarms, have been removed, as shown in Table III.

In Tables II and III, "Number" denotes the number of records, while "Label Frame Size" suggests that when a fall event occurs, range-Doppler images from frames 17 to 24 are labeled as 1, with the remaining frames labeled as 0. The length of each record is randomized. Due to the varying recording times of each record and the presence of nonfall frames in the fall data, the ratio of fall to nonfall frames is not consistent across the datasets. However, when we sampled 10% of the datasets for calculation, we observed that the average ratio of fall to nonfall frames in Dataset A was approximately 1:43, while in Dataset B, the average ratio was approximately 1:20.

D. Evaluation of the Trigger Model

Given the limited data available, we maintained a ratio of 9:1 between the training set and the validation set. The trigger model is predominantly used to anticipate scenarios with significant changes in speed and distance rather than to detect whether a fall has occurred. Consequently, the selection of the appropriate dataset for building the model is crucial to effectively assess its performance.

In addition to collecting fall data for each field, our datasets included a significant amount of nonfall data. To ensure that the trigger model is capable of detecting most fall incidents, it was necessary to build the model using all the available fall data. However, some nonfall data exhibited substantial variations in terms of speed and distance, which could potentially affect the accuracy of the fall detection model. To address this issue, we divided the collected data into two datasets for building the trigger model, namely, Dataset A and Dataset B, and compared their respective performance using a confusion matrix [29].

TABLE IV
EVALUATION METRICS OF THE TRIGGER MODEL RESULT

	Accuracy	Recall	Precision
Dataset A	92.20%	95.47%	51.77%
Dataset B	91.21%	91.60%	61.54%

Table IV presents the experimental result of the trigger model, indicating an accuracy and recall of over 90% for both datasets. However, there is a notable difference in precision of almost 10%. The trigger model performed well in distinguishing fall and nonfall events by detecting significant changes in speed and distance from range-Doppler image.

To understand the difference in precision, we analyzed the data and found that many nonfall events exhibit similar changes in speed and distance as fall incidents. As a result, the model tends to classify nonfall events as fall, leading to a decrease in precision as the trigger range increases.

E. Evaluation of the Fall Detection Model

The fall detection model has been trained with a training-to-validation set ratio of 9:1. In a previous experiment, we discovered that the trigger model performed more effectively when trained on Dataset B. However, relying solely on either of these separate datasets did not ensure optimal results. To overcome this limitation, we propose two different dataset combinations for model building within the system. In the current experiment, we have divided the datasets according to these proposed combinations, as detailed in the following.

1) Combination A: Both the trigger model and the fall detection model are trained using Dataset A.

2) Combination B: The trigger model is trained using dataset B and the fall detection model is trained with Dataset A.

The experimental results, as shown in Table V, indicate that the trigger model was trained on Dataset B, while the fall detection model was trained on Dataset A. When these two models are combined, they yield all evaluation metrics of over 90%, achieving a precision that is over 12% higher than that of Combination A.

The labeling processes for both the trigger model and the fall detection model are identical, except for differences in the nonfall data used in model training. Our findings suggest that using excessive amounts of data that show marked changes in speed and distance, but not actual falls, does not lead to effective training of the trigger model. Therefore, the fall detection model is tasked with predicting data that could potentially result in false alarms, whereas the trigger model is specifically designed to predict data pertaining to possible falls.

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TABLE V
EVALUATION METRICS OF THE FALL DETECTION MODEL RESULT

Combination	Accuracy	Recall	Precision
A	92.73%	91.43%	78.05%
B	96.82%	90.91%	90.91%

TABLE VI
RESULT OF THE ADDITIONAL EXPERIMENT

	Fall Data (Correct / Total)	Non-Fall Data (Correct / Total)
Office	82/85	98/107
Bathroom 1	22/22	5/5
Bathroom 2	9/9	15/15
Bathroom 3	14/14	13/14
Other Office	4/5	
Hold the shower		26/30
Total	131/135	157/171

TABLE VII
EVALUATION METRICS OF THE ADDITIONAL EXPERIMENTAL RESULT

Accuracy	Recall	Precision	F1-Score
94.12%	97.03%	90.34%	93.90%

F. Additional Experiment

We conducted an additional experiment to complete the design of the fall detection system because the validation set in the original dataset had a limited amount of data. This experiment used additional data that we collected and is not included in the original dataset.

To collect data for this experiment, we randomly performed irregular fall behavior and nonfall behavior in different fields. The fall behavior is captured by counting the number of times it occurs, whereas nonfall behavior is captured every 10 s. There are around five to seven participants.

Tables VI and VII present the results of the additional experiment. The findings demonstrate that the recall rate is high, exceeding 97%, which indicates that the probability of misclassifying a fall event as a nonfall event is minimal. However, the precision rate is not as high due to the abundance of nonfall actions, resulting in a higher probability of false alarms. Nonetheless, in fall detection scenarios, it is preferable to risk a false alarm by misclassifying a non-fall event as a fall, rather than failing to detect an actual fall. Overall, each evaluation metric exceeds 90%.

G. Model Complexity Comparison

Ou and Ye [30] proposed a lightweight deep learning model for radar-based fall detection, which has around 3 K parameters. The model was compared with other classical models, such as Mobile-RadarNet [31] (14 K parameters) and DivNet [32] (5.6 M parameters), showing a significant reduction in parameter count while maintaining a certain level of accuracy. Although the model proposed by [30] can significantly reduce computational resources, the possibility of false alarms needs to be improved. Hence, in our research, a novel inference mechanism was proposed, namely, a more complex model inference is required only when the range-Doppler image is continuously triggered by the proposed trigger model. Implementing this approach not only mitigates the model's complexity but also attenuates the likelihood of false alarms.

Given that the time of a fall event in daily life is minimal, the aim of this study is to reduce the complexity of the proposed trigger model, avoiding the need for large computational and memory resources at each time interval. This allows for potential implementation on the Internet of Things (IoT) devices. According to the experimental results of [30], in most cases, fewer model parameters can reduce inference time and memory resources. The total parameter count of the proposed trigger model in this research is approximately 2 K. Combined with the proposed inference mechanism, it can further reduce the possibility of false alarms, as evidenced by the precision of Tables IV and V.

V. CONCLUSION

In this study, we proposed a novel range-Doppler preprocessing method and two deep learning models for radar-based fall detection. Our trigger model can reduce computational complexity, while the fall detection model considers continuous falling behavior, resulting in high experimental performance with evaluation metrics exceeding 90% and a recall rate of 97% in an additional experiment. While the proposed system is adaptable to different indoor environments and achieves high accuracy, some limitations and challenges, such as the impact of environmental factors on the system's performance, need to be addressed.

The practical implications of our findings suggest that the proposed system holds promise for use in assisted living facilities, hospitals, and other settings, where fall detection is critical. Future research should focus on the collection of more data to further improve the system's accuracy, as well as the refinement of the deep learning models to enhance their performance. Our study underscores the feasibility and effectiveness of our novel methods specifically tailored for radar-based fall detection, laying a solid foundation for future research in this area.

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